Thomas Nikolaus

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Research Interests

Higher-categorical methods in algebra, geometry and topology, particularly in algebraic K-theory, homotopy theory, and higher algebra, with connections to mathematical physics and noncommutative geometry.

My current work focuses on trace methods, prismatic cohomology, continuous K-theory, equivariant homotopy theory and assembly conjectures.

Selected Publications

These papers reflect my main research directions and include some of my most significant results.

- Hermitian K-theory for stable ∞-categories III (with B. Calmès, E. Dotto, Y. Harpaz, F. Hebestreit, M. Land, K. Moi, D. Nardin and W. Steimle), *Annals of Mathematics* 204 (2026), no. 1, 119–220.
- Maps between spherical group rings (with S. Carmeli and A. Yuan), to appear in *Inventiones Mathematicae*, 2026.
- Hermitian K-theory for stable ∞-categories II (with B. Calmès, E. Dotto, Y. Harpaz, F. Hebestreit, M. Land, K. Moi, D. Nardin and W. Steimle), *Acta Mathematica* 235 (2025), 149–400.
- On the K-theory of $\mathbb{Z}/p^n$ (with B. Antieau and A. Krause), preprint, arXiv:2405.04329.
- L-theory of C^*-algebras (with M. Land, M. Schlichting), *Proceedings of the LMS* 127 (2023), 1451–1506.
- On the Beilinson fiber square (with B. Antieau, A. Mathew, M. Morrow), *Duke Mathematical Journal* 171 (2022), 3707–3806.
- Frobenius homomorphisms in higher algebra, *Proceedings of the International Congress of Mathematicians 2022*.
- Cartier modules and cyclotomic spectra (with B. Antieau), *Journal of the AMS* 34 (2021), 1–78.
- The Balmer spectrum of the equivariant homotopy category of a finite abelian group (with T. Barthel, M. Hausmann, N. Naumann, J. Noel, N. Stapleton), *Inventiones Mathematicae* 216 (2019), 215–240.
- On topological cyclic homology (with P. Scholze), *Acta Mathematica* 221 (2018), 203–409.

Academic Leadership, Funding, and Service

- Since 2026: Spokesperson of the Centre for Mathematics Münster (CMM).
- Since 2024: Spokesperson of the Cluster of Excellence EXC 2044 “Mathematics Münster: Dynamics – Geometry – Structure” (approx. 200 members). Co-led the successful renewal application for the second funding period, 2026–2032.
- 2021: Together with Mario Ohlberger responsible for the successful application of the Centre for Mathematics Münster (CMM) as a 41-million-euro research-building project under the German *Forschungsbauten* programme pursuant to Art. 91b GG.
- Since 2026: Co-organiser of AI.MATH, an initiative developing positions on the future of mathematical research and mathematics degree programmes in the age of AI.
- Scientific host and mentor for the independently funded DFG projects of Devarshi Mukherjee (Research Grant 534946574, including a Temporary Position for Princi-

pal Investigator) and Hyungseop Kim (Walter Benjamin Programme 576038862); involvement in both successful proposals.
- Since 2020: Principal Investigator in CRC 1442 “Geometry: Deformations and Rigidity”.
- Since 2019: Principal Investigator in the Cluster of Excellence “Mathematics Münster: Dynamics - Geometry - Structure”.
- 2014–2018: Principal Investigator in SFB 1085 “Higher Invariants - Interactions between Arithmetic Geometry and Global Analysis”, Regensburg.
- 2012–2015: Main applicant of the DFG research network “String Geometry”.
- 2024–2025: Member of the Topology Section Panel for ICM 2026, responsible for recommending invited sectional speakers.
- 2023–2026: Editorial Board, *Journal of the European Mathematical Society*.
- Since 2021: Member of the K-Theory Foundation.
- 2020–2024: Editorial Board, *Algebraic & Geometric Topology*.
- 2021–2024: Editor of *Manifolds and K-theory: Memorial volume in honour of Andrew Ranicki*.

Positions

since 2018	Professor, University of Münster
2017–2018	Nachwuchsgruppenleiter (Research Group Leader), W2, Max Planck Institute for Mathematics, Bonn
2015–2016	Postdoctoral Researcher, Max Planck Institute for Mathematics, Bonn
2014–2015	Vertretungsprofessor (Acting Professor), University of Bonn
2011–2014	Akademischer Rat auf Zeit (postdoctoral position, fixed-term), University of Regensburg
2010–2011	Research Associate, Landesexzellenzcluster “Connecting Particles with the Cosmos”, University of Hamburg
2009–2010	Research Associate, DFG Collaborative Research Centre 676 “Particles, Strings, and the Early Universe”, University of Hamburg

Academic Distinctions

- 2024: von Kaven-Ehrenpreis für Mathematik.
- 2023: Frontiers of Science Award at the ICBS.
- 2022: Sectional talk at the International Congress of Mathematicians.
- 2006–2009: Scholar of the German National Academic Foundation.

Selected Invited Lectures and Lecture Series

- 2026: Mordell Lecture, University of Cambridge.
- 2025: Course, Luis Santaló School / CIMPA, Buenos Aires.
- 2024: Hausdorff Colloquium, University of Bonn.
- 2024: MATH+ Friday Colloquium, Berlin.
- 2024: Lecture Series, Masterclass on Continuous K-Theory, University of Copenhagen.
- 2023: Research Lecture, IHES Summer School on Algebraic K-Theory.
- 2022: Zurich Colloquium in Mathematics, ETH Zürich / University of Zurich.
- 2018: Main Lecture Series, Isaac Newton Institute, Cambridge.
- 2017: Mathematics Colloquium, University of California, Berkeley.
- 2017: Lecture Series, Masterclass on Stable Homotopy Theory and p-adic Hodge Theory, University of Copenhagen.

Supervision

Current PhD Students

- Thorger Geiss: Ad-theories and (∞, ∞)-spectra.
- Edith Hübner: Spherical δ-rings
- Phil Pützstück: Condensed Anderson Duality.

Completed PhD Theses

- Stefano Ariotta: *Coherent cochain complexes and Beilinson t-structures*, University of Münster, 2022. Now Backend Engineer at DeepL.
- Jonas McCandless: *TR and its relation to algebraic K-theory*, University of Münster, 2022. After a postdoc at MPIM Bonn, now Software Developer in Reality Capture at Dalux.
- Marin Janssen: *Genuine cohomology theories*, University of Münster, 2024. Now at PICTURE GmbH.
- Konrad Bals: *On arithmetic cohomology and topological periodic theories*, University of Münster, 2025. Now working in agricultural technology R&D at NEXAT.

Master's and Bachelor's Theses

Master's theses: 15 completed; currently supervising 5 further projects. Bachelor's theses: several completed.

Current Postdocs

- Dr. Anish Chedalavada, since 2026.
- Dr. Georg Lehner, since 2025.
- Dr. Lucas Piessevaux, since 2026.
- Dr. Maxime Ramzi, since 2024.
- Dr. Catherine Ray, since 2023.
- Dr. Florian Riedel, since 2026.
- Dr. Ferdinand Wagner, since 2026.
- Dr. Hyungseop Kim, incoming March 2027 (DFG Walter Benjamin Fellow).

Former Postdocs

- Dr. Devarshi Mukherjee (2024–2025), now Lecturer at the University of Aberdeen.
- Dr. Achim Krause (2018–2024), now Professor at the University of Oslo.
- Dr. Jay Shah (2021–2024), now Research Scientist in accelerated computing at Colfax International.
- Dr. Julia Semikina (2020–2023), now Associate Professor at the University of Lille.
- Prof. Dr. Manuel Krannich (2021–2022), now Professor at Karlsruhe Institute of Technology.
- Prof. Dr. Fabian Hebestreit (2021–2022), now Professor at Bielefeld University.
- Dr. Christoph Schrade (2018–2021), now Research Associate in Information Management at Leipzig University.

Education

2011 PhD in Mathematics, University of Hamburg. Advisor: Christoph Schweigert. Thesis: "Higher Categorical Structures in Geometry - General Theory and Applications to Quantum Field Theory"

2009 Diploma in Mathematics, University of Hamburg. Advisor: Christoph Schweigert. Thesis: "Äquivariante Gerben und Abstieg"

Conference, Seminar, and Workshop Organisation

- Organiser, "Categories and Symmetries – a conference in honour of Christoph Schweigert's 60th birthday", 23–27 August 2027.
- Organiser, Gauß Lecture 2027, with speaker Catharina Stroppel, Münster, 12 May 2027.
- Organiser, "AI and the Future of Mathematics", an online seminar on the impact of artificial intelligence on mathematics, from October 2026.
- Organiser, Opening Conference for the second funding period of Mathematics Münster, Münster, 16–18 September 2026.
- Organiser, Beyond the telescope conjecture, Cambridge, June 2025.
- Organiser, Young Topologists Meeting, Münster, August 2024.
- Organiser, "From Homotopy Theory to Analysis - a conference in honor of Ulrich Bunke's 60th Birthday", Greifswald, March 2024.
- Organiser, Mid-term Conference Mathematics Münster, March 2023.
- Organiser, Oberwolfach Workshop on Non-commutative Geometry, August 2022.
- Organiser, ICM Sectional Workshop, Copenhagen, July 2022.
- Organiser, Oberwolfach Arbeitsgemeinschaft, April 2019.
- Organiser, European Autumn School in Topology, Utrecht, September 2017.
- Organiser, Conference on Invertible Objects and Duality in Derived Algebraic Geometry and Homotopy Theory, Regensburg, April 2017.
- Organiser, European Autumn School in Topology, Utrecht, September 2016.
- Organiser, "Modular Invariants in Topology and Analysis", Regensburg, September 2014.
- Organiser, "From Poisson to String Geometry", Erlangen, September 2012.

Publications

1. Unfolding of symmetric monoidal (∞, n)-categories (with R. Haugseng). Preprint, arXiv:2611.01994.
2. A short proof of the Segal conjecture for C_p. Preprint, arXiv:2610.39440.
3. Unbounded Weight Structures: (Re)construction and Completion (with P. Pützstück). Preprint, arXiv:2605.00783.
4. Bredon sheaf cohomology (with G. Arnone and D. Mukherjee). Preprint, arXiv:2604.08066.
5. Dualisable categories in noncommutative geometry (with D. Mukherjee). To appear in *Lecture Notes on K-theory and Operator Algebra*, EMS Press.
6. Trace methods for stable categories I: The linear approximation of algebraic K-theory (with Y. Harpaz and V. Saunier). Preprint, arXiv:2411.04743.
7. An alternative to spherical Witt vectors (with M. Yakerson). To appear in *Proceedings of the American Mathematical Society*. arXiv:2405.09606.
8. Maps between spherical group rings (with S. Carmeli and A. Yuan). To appear in *Inventiones Mathematicae*. arXiv:2405.06448
9. On the K-theory of $\mathbb{Z}/p^n$ (with B. Antieau and A. Krause). Preprint, arXiv:2405.04329.
10. Witt vectors with coefficients and TR (with E. Dotto, A. Krause and I. Patchkoria). *Proceedings of the London Mathematical Society* 130 (2025), no. 5, e70047. arXiv:2312.12971.
11. Prismatic cohomology relative to δ-rings (with B. Antieau and A. Krause). To appear in Annales scientifiques de l'École normale supérieure. arXiv:2310.12770.
12. Polygonic spectra and TR with coefficients (with A. Krause and J. McCandless). Preprint, arXiv:2302.07686.
13. L-theory of C^*-algebras (with M. Land and M. Schlichting). *Proceedings of the London Mathematical Society* 127 (2023), no. 5, 1451–1506. arXiv:2208.10556.
14. Frobenius homomorphisms in higher algebra. In *Proceedings of the International Congress of Mathematicians 2022*.

15. On the K-theory of $\mathbb{Z}/p^n$ – announcement (with B. Antieau and A. Krause). Preprint, arXiv:2204.03420.
16. K-theory of $\mathbb{Z}/p^k$ and prismatic cohomology. Oberwolfach contribution, Report No. 34/2022, p. 1979.
17. K-theory and polynomial functors (with C. Barwick, S. Glasman and A. Mathew). Preprint, arXiv:2102.00936.
18. Hermitian K-theory for stable ∞-categories I: Foundations (with B. Calmès, E. Dotto, Y. Harpaz, F. Hebestreit, M. Land, K. Moi, D. Nardin and W. Steimle). *Selecta Mathematica, New Series* 29 (2023), no. 1, Paper No. 10. arXiv:2009.07223.
19. Hermitian K-theory for stable ∞-categories II: Cobordism categories and additivity (with B. Calmès, E. Dotto, Y. Harpaz, F. Hebestreit, M. Land, K. Moi, D. Nardin and W. Steimle). *Acta Mathematica* 235 (2025), no. 2, 149–400. arXiv:2009.07224.
20. Hermitian K-theory for stable ∞-categories III: Grothendieck–Witt groups of rings (with B. Calmès, E. Dotto, Y. Harpaz, F. Hebestreit, M. Land, K. Moi, D. Nardin and W. Steimle). Annals of Mathematics 204 (2026), no. 1, 119–220. arXiv:2009.07225.
21. On the homotopy type of L-spectra of the integers (with F. Hebestreit and M. Land). *Journal of Topology* 14 (2021), 183–214. arXiv:2004.06889.
22. The initial quadratic form on an abelian group (with F. Hebestreit and M. Land). Short note/erratum to a remark in "On the homotopy type of L-spectra of the integers". https://www.uni-muenster.de/IVV5WS/WebHop/user/nikolaus/papers.html.
23. On the Beilinson fiber square (with B. Antieau, A. Mathew and M. Morrow). *Duke Mathematical Journal* 171 (2022), no. 18, 3707–3806. arXiv:2003.12541.
24. Witt vectors with coefficients and characteristic polynomials over non-commutative rings (with E. Dotto, A. Krause and I. Patchkoria). *Compositio Mathematica* 158 (2022), 366–408. arXiv:2002.01538.
25. Bökstedt periodicity and quotients of DVRs (with A. Krause). *Compositio Mathematica* 158 (2022), 1683–1712. arXiv:1907.03477.
26. Beilinson's fibre sequence in K-theory via topological cyclic homology. Oberwolfach contribution, Report No. 36/2019, p. 2237.
27. Bökstedt periodicity and quotients of DVRs (joint work with A. Krause). Oberwolfach contribution, Report No. 29/2019, 1757–1759.
28. Topological cyclic homology (with L. Hesselholt). In *Handbook of Homotopy Theory*, CRC Press/Chapman Hall Handbooks in Mathematics, 619–656, 2020. arXiv:1905.08984.
29. Algebraic K-theory of planar cuspidal curves (with L. Hesselholt). In *K-theory in Algebra, Analysis and Topology*, Contemporary Mathematics 749, American Mathematical Society, 139–148, 2020. arXiv:1903.08295.
30. Cartier modules and cyclotomic spectra (with B. Antieau). *Journal of the American Mathematical Society* 34 (2021), 1–78. arXiv:1809.01714.
31. Higher geometry for non-geometric T-duals (with K. Waldorf). *Communications in Mathematical Physics* 374 (2020), 317–366. arXiv:1804.00677.
32. Topological Hochschild homology of stable ∞-categories. Oberwolfach contribution, Report No. 15/2018, 857–865.
33. The Balmer spectrum of the equivariant homotopy category of a finite abelian group (with T. Barthel, M. Hausmann, N. Naumann, J. Noel and N. Stapleton). *Inventiones Mathematicae* 216 (2019), 215–240. arXiv:1709.04828.
34. On the Blumberg–Mandell Künneth theorem for TP (with A. Mathew and B. Antieau). *Selecta Mathematica, New Series* 24 (2018), 4555–4576. arXiv:1710.05658.
35. On topological cyclic homology (with P. Scholze). *Acta Mathematica* 221 (2018), 203–409. arXiv:1707.01799.
36. Correction to "On topological cyclic homology" (with P. Scholze). *Acta Mathematica* 222 (2019), 215–218.
37. Localization of cofibration categories and groupoid C^*-algebras (with M. Land and K. Szumiło). *Algebraic & Geometric Topology* 17 (2017), 3007–3020. arXiv:1609.03805.

38. On the relation between K- and L-theory of C^*-algebras (with M. Land). *Mathematische Annalen* 371 (2018), 517–563. arXiv:1608.02903.
39. Stable ∞-operads and the multiplicative Yoneda lemma. Preprint, arXiv:1608.02901.
40. Cyclotomic Structures and Factorization Homology (joint work with P. Scholze). Oberwolfach contribution, Report No. 35/2016, 2021–2022.
41. The Beilinson regulator is a map of ring spectra (with U. Bunke and G. Tamme). *Advances in Mathematics* 333 (2018), 41–86. arXiv:1509.05667.
42. Homology of dendroidal sets (with M. Bašić). Preprint, arXiv:1509.00702.
43. Presentably symmetric monoidal ∞-categories are represented by symmetric monoidal model categories (with S. Sagave). *Algebraic & Geometric Topology* 17 (2017), 3189–3212. arXiv:1506.01475.
44. Lax colimits and free fibrations in ∞-categories (with D. Gepner and R. Haugseng). *Documenta Mathematica* 22 (2017), 1225–1266. arXiv:1501.02161.
45. The Universal Property of Global Spectra and Elliptic Cohomology (joint work with D. Gepner). Oberwolfach contribution, Report No. 14/2015, 759–761.
46. Twisted differential cohomology (with U. Bunke). *Algebraic & Geometric Topology* 19 (2019), 1631–1710. arXiv:1406.3231.
47. Universality of multiplicative infinite loop space machines (with D. Gepner and M. Groth). *Algebraic & Geometric Topology* 15 (2015), 3107–3153. arXiv:1305.4550.
48. T-duality via gerby geometry and reductions (with U. Bunke). *Reviews in Mathematical Physics* 27 (2015), no. 5, 1550013, 46 pp. arXiv:1305.6050.
49. Algebraic K-theory of ∞-operads.
Journal of K-Theory 14 (2014), no. 3, 614–641. arXiv:1303.2198.
50. Principal ∞-bundles: Presentations (with U. Schreiber and D. Stevenson). *Journal of Homotopy and Related Structures* 10 (2015), 565–622. arXiv:1207.0249.
51. Principal ∞-bundles: General theory (with U. Schreiber and D. Stevenson). *Journal of Homotopy and Related Structures* 10 (2015), 749–801. arXiv:1207.0248.
52. Differential cohomology theories as sheaves of spectra (with U. Bunke and M. Völkl). *Journal of Homotopy and Related Structures*. arXiv:1311.3188.
53. Dendroidal sets as models for connective spectra (with M. Bašić). *Journal of K-Theory* 14 (2014), no. 3, 387–421. arXiv:1203.6891.
54. Lifting problems and transgression for non-abelian gerbes (with K. Waldorf). *Advances in Mathematics* 242 (2013), 50–79. arXiv:1112.4702.
55. Four equivalent versions of non-abelian gerbes (with K. Waldorf). *Pacific Journal of Mathematics* 264 (2013), no. 2, 355–420. arXiv:1103.4815.
56. Bicategories in field theories – an invitation (with C. Schweigert). In *Strings, Gauge Fields, and the Geometry Behind*, World Scientific, 119–132, 2013. arXiv:1111.6896.
57. A smooth model for the string group (with C. Sachse and C. Wockel). *International Mathematics Research Notices* 2013, no. 16, 3678–3721. arXiv:1104.4288.
58. Strictification of weakly equivariant Hopf algebras (with J. Maier and C. Schweigert). *Bulletin of the Belgian Mathematical Society – Simon Stevin* 20 (2013), no. 2, 269–285. arXiv:1109.0236.
59. Equivariant modular categories via Dijkgraaf–Witten theory (with J. Maier and C. Schweigert). *Advances in Theoretical and Mathematical Physics* 16 (2012), no. 1, 289–358. arXiv:1103.2963.
60. Equivariance in higher geometry (with C. Schweigert). *Advances in Mathematics* 226 (2011), no. 4, 3367–3408. arXiv:1004.4558.
61. Algebraic models for higher categories. *Indagationes Mathematicae, New Series* 21 (2011), no. 1–2, 52–75. arXiv:1003.1342.
62. Algebraic models for higher categories. Oberwolfach contribution, Report No. 25/2010, 1523–1524.
63. Bundle gerbes and surface holonomy (with J. Fuchs, C. Schweigert and K. Waldorf). In *European Congress of Mathematics*, EMS Publishing House, 167–197, 2008. arXiv:0901.2085.